

The Climate Tax of AI: What Google's Data Centers Reveal About the Hidden Cost of AI

This article covers three things: 1) how data centers connect to energy and emissions, 2) what the actual numbers look like across Google's facilities, and 3) a simple regression based model to estimate efficiency in growing markets.

I saw a meme on TikTok last week: "I didn't drink any water today so yall can use ChatGPT. You're welcome". I laughed – and then I couldn't stop thinking about it.

I went down a rabbit hole on the environmental cost of running data centers. And guess what? Water is not the primary environmental constraint. All U.S. data centers combined are estimated to use tens of billions of gallons—roughly 1–2% water usage of golf irrigation and 0.1% of corn agriculture. The bigger environmental driver is electricity. Data centers run 24/7, and their carbon footprint scales directly with grid intensity and total power demand. That led me to a bigger question:

How much carbon are we actually emitting to power all this AI?

AI infrastructure has been growing at a pace that's hard to grasp. Between 2019 and 2024, the number of hyperscale data centers in the world doubled. By 2030, the sector is expected to double in size again. And the four largest tech companies plan to spend over \$370 billion on data center infrastructure in 2025 alone.

So how much power does it actually take to run one of these things?

A single hyperscale data center typically runs at around 100 megawatts of power – roughly enough to power about 80,000 American homes, running 24/7 for the whole year.

And that's just to keep the servers running. On top of that, you need additional energy to keep them cool. Think of it like your laptop overheating but at the scale of several football fields. To keep the servers from overheating, you need two things: electricity and cooling. A lot of both.

Why can't they just build data centers in cold regions?

Two reasons: latency and cost. If your customers are in Southeast Asia, serving them from a data center in Helsinki adds about 200–300 milliseconds of delay. That might not sound like much, but for AI applications that are supposed to feel instant, that difference is noticeable. Although Helsinki is cold and requires less cooling energy, electricity cost in Helsinki is about four times more expensive than in Jakarta.

Is Jakarta a better option then?

From a cost perspective, yes. From the environmental perspective? Likely not. Here's why: Every time a data center uses electricity, it indirectly creates carbon emissions – that's the CO₂ released when energy is produced, especially from burning fuels. In many places, those emissions have a real cost, even if it's not always paid directly.

Where that electricity comes from matters a lot. Some energy sources are clean, like wind, solar, hydro, and nuclear – they generate power with little to no carbon emissions, but aren't equally available everywhere. Others are dirtier, like coal, oil, and natural gas, because they release CO₂ when burned. If you want a quick refresher on how the carbon cycle works, I've linked a short video [here](#).

Indonesia's grid runs about 60% on coal, which means most of the electricity powering a Jakarta data center comes from burning fossil fuels. In contrast, Finland relies heavily on nuclear and hydro, which are much cleaner. Same servers, same workload – completely different carbon footprint, just based on where they're located.

And demand is making this worse. As data center usage surges, some utilities are even bringing coal plants back online just to keep up.

If AI is really that damaging to the environment, can't we just ask these companies to stop investing in AI?

Probably not. These companies exist to **deliver shareholder value**. If there's enough demand for AI, they'll build the infrastructure to meet it. What they can control is how efficiently they use the energy they consume. Let's go back to the laptop analogy. A newer gen laptop with a fancy processor doesn't heat up as fast, even when running multiple apps. Data centers work the same way. Better hardware, smarter cooling systems, and favorable climate all reduce the energy wasted on overhead rather than actual computing. So the real question isn't whether to build. It's how efficiently you build and where.

Which leads us to: *how efficient are tech companies actually being?*

To answer that, we need one metric: PUE, or Power Usage Effectiveness. The formula is simple – total facility power (all power used by the building) divided by IT equipment power (power used by the servers themselves). A PUE of 1.0 would mean every watt entering the building goes directly to computing. Nothing wasted on cooling or overhead. In practice that's impossible, but the closer to 1.0 the better. A PUE of 1.09 means for every dollar spent powering servers, an extra 9 cents goes facility overhead – cooling, power distribution, lighting, and losses. Here's how the three major hyperscalers compare:

Company	2024 Global PUE	Granularity
Google	1.09	Facility-level
AWS	1.15	Global average only
Microsoft	1.16	Regional

One thing I want to call out is that PUE comes with real limitations. Facility age, cooling technology, and utilization rate all influence the number in ways that have nothing to do with how efficiently a company actually operates. A data center running at 30% usage capacity will report a worse PUE than one running at 90%, because the cooling infrastructure is still running at full overhead regardless of how many servers are actually doing work. BUT comparing PUE across hyperscalers isn't just about the number. The transparency gap says a lot about how seriously these companies take their net zero commitments.

Wait... what has the government been doing to control this?

The EU is one of the few regions with a formal Code of Conduct that uses PUE as a benchmark metric and sets best-practice targets. One of the policies is ISO/IEC 30134-2 which defines how PUE should be measured and reported so that comparisons can be meaningful.

From an environmental cost perspective, the EU Emissions Trading System (ETS) works like a cap-and-trade program. The EU sets a limit on total emissions, companies buy permits to pollute, and the market determines the price. Right now that price is \$70.37 per ton of CO₂. That's still below the true cost.

Economists estimate the true damage from emitting 1 ton of CO₂ – measured as the Social Cost of Carbon (SCC) which is basically the present value of future impacts like heat damage, sea-level rise, and health costs – at \$185/ton (EPA 2023, Resources for the Future) to \$283/ton (PNAS, December 2024).

What about other countries outside of the EU? How much carbon tax do they charge?

Here's a quick comparison between what EU ETS charges vs. the federal government across different countries.

Country	Carbon Price (\$ / ton)	Status vs. EU ETS (\$70.37)
Denmark	\$178.80	Above (national tax + EU ETS)
Netherlands	\$165.20	Above
Ireland	\$138.87	Above
Finland	\$137.26	Above
Belgium / Poland	\$70.37	At benchmark (EU ETS only)
Australia	\$21.82	Below
Singapore	\$18.62	Below
South Korea	\$6.45	Below
Japan	\$1.93	Below

Indonesia	\$0.72	Far below
USA	\$0.00	No carbon price at Federal level, but states like California and Washington charge ~\$26 per ton
Malaysia / Thailand India / Saudi Arabia / UAE / Taiwan	\$0.00	No carbon price

Source: World Bank Carbon Pricing Dashboard.

These carbon prices apply across industries. It's not a direct "bill" charged to data centers, but the cost shows up indirectly – mainly through electricity prices.

How much that cost depends heavily on the country. Indonesia, for example, is the world's largest coal exporter, and roughly 60% of its electricity comes from coal. A high carbon price would directly impact the economics of that industry, which is why carbon pricing remains relatively low.

To address this kind of imbalance, the European Union introduced the Carbon Border Adjustment Mechanism (CBAM). The idea is simple: if emissions happen in a low-carbon-price country but the product is consumed in Europe, the importer pays the difference at the border. This currently applies to physical goods like steel, cement, and aluminum.

A data center in Jakarta serving European AI users is, in effect, doing what CBAM was designed to prevent. But for now, there's no adjustment, because CBAM doesn't apply to digital services.

So what if there were? If a CBAM-style mechanism applied to data centers, how much would that actually cost?

To estimate this, I used Google's publicly available PUE data across 30 facilities from 2022–2025, and assumed a 100 MW IT load facility (the typical scale for a hyperscale campus). Using PUE data from 10 selected facilities and the EU ETS as a benchmark for carbon pricing, I estimated:

1. Cooling cost based on electricity prices,
2. CO₂ emissions using grid carbon intensity, and
3. Carbon cost under a consistent pricing framework.

Google Facility (Actual 2025 Avg PUE)	2025 Avg PUE	Cooling Overhead (MW)	Cooling Overhead (kWh/yr)	Electricity Price (\$/kWh)	Annual Cooling Cost	Grid Carbon Intensity (gCO ₂ /kWh)	Primary Energy Source (2024, % of grid)**	Annual CO ₂ from Cooling (tons)	Local Carbon Price (\$/ton)	Current Carbon Cost***	Carbon Cost Gap (Current vs EU ETS)
● FAVORABLE — Low PUE · Clean grid · At or above EU ETS carbon price											
Fredericia, Denmark	1.062	6.2 MW	54.8M kWh	\$0.234	\$12.8M	155	Wind 58%, Solar 14%, Bioenergy 16%	8,486	\$178.80	\$1.52M	+\$920K
Eemshaven, Netherlands	1.073	7.3 MW	63.5M kWh	\$0 GW.220	\$14.0M	253	Natural Gas 40%, Wind 30%, Solar 14%	16,081	\$165.20	\$2.66M	+\$1.53M
Dublin, Ireland	1.080	8.0 MW	70.1M kWh	\$0.270	\$18.9M	256	Natural Gas 50%, Wind 35%, Solar 5%	17,933	\$138.87	\$2.49M	+\$1.23M
Hamina, Finland	1.093	9.3 MW	81.0M kWh	\$0.124	\$10.0M	57	Nuclear 35%, Hydro 28%, Wind 24%	4,603	\$137.26	\$632K	+\$308K
■ MODERATE — Mid PUE · Dirty grid · No carbon price (US)											
Midlothian, Texas	1.103	10.3 MW	89.8M kWh	\$0.148	N/A*	370	Natural Gas 43%, Wind 26%, Coal 15%	33,222	\$0.00	\$0	-\$2.34M
Council Bluffs, Iowa	1.110	11.0 MW	96.4M kWh	\$0.148	N/A*	286	Wind 58%, Natural Gas 20%, Nuclear 10%	27,559	\$0.00	\$0	-\$1.94M
● MATERIAL GAP — Higher PUE · Dirty grid · Symbolic or zero carbon price											
Inzai, Japan	1.113	11.3 MW	98.6M kWh	\$0.202	\$19.9M	483	LNG 35%, Coal 30%, Nuclear 10%	47,639	\$1.93	\$92K	-\$3.26M
Mayes County, Oklahoma	1.123	12.3 MW	107.3M kWh	\$0.148	N/A*	297	Natural Gas 40%, Wind 35%, Coal 15%	31,871	\$0.00	\$0	-\$2.24M
● LARGEST GAP — High PUE · Dirty grid · Near-zero or zero carbon price											
Changhua County, Taiwan	1.130	13.0 MW	113.9M kWh	\$0.187	\$21.3M	635	Coal 45%, LNG 35%, Nuclear 10%	72,337	\$0.00	\$0	-\$5.09M
Singapore (2nd facility)	1.145	14.5 MW	127.0M kWh	\$0.265	\$33.7M	499	Natural Gas 95%, Solar 3%	63,345	\$18.62	\$1.18M	-\$3.28M

*US commercial electricity prices vary by state and are not in the GlobalPetrolPrices national average dataset.

**Primary energy source percentages are approximate 2024 national grid averages (Ember / IEA). Actual power mix at each facility may differ depending on PPAs and local grid conditions. Google purchases renewable energy certificates and PPAs that may offset grid-average carbon intensity.

***Carbon costs represent what a facility would owe if carbon were priced directly on cooling emissions at local rates — not what appears on electricity bills. No jurisdiction currently charges data centers a direct per-ton carbon fee.

Hamina, Finland is basically what “good” looks like. Google operates at a PUE of 1.093 on a grid with ~57 gCO₂/kWh carbon intensity. To put that in context, anything below 100 is considered clean. Natural gas typically lands around 300–500, and coal-heavy grids can go above 600. Finland sits way below all of that.

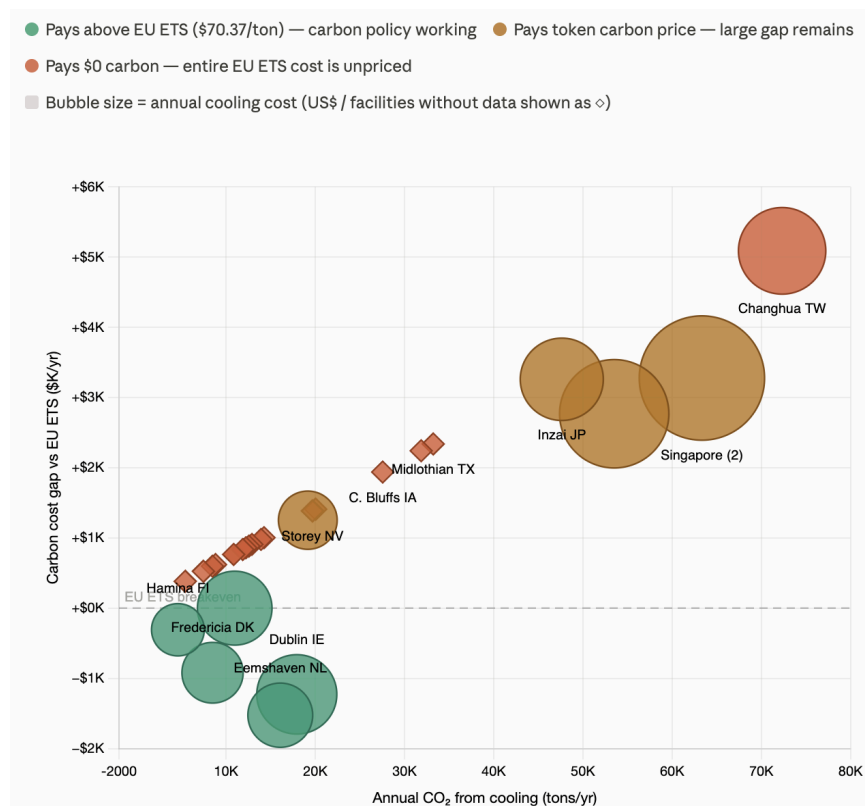
That difference shows up quickly in the numbers. At ~\$137/ton, the implied carbon cost is already higher than the EU ETS benchmark. A big reason for that is Finland’s energy mix—lots of nuclear and renewables—combined with consistent policy and long-term investment.

Texas (Midlothian) is where things start to look very different. The PUE is almost the same at 1.103, but the grid is not. At ~370 gCO₂/kWh, it’s squarely in natural gas territory—roughly 6x higher than Finland. That gap alone drives most of the emissions difference.

It adds up to about \$2.34M a year in carbon cost that, in practice, isn’t really being paid today. The United States doesn’t have a federal carbon tax, so instead of charging for emissions, it mostly relies on subsidies and incentives to shift behavior.

Inzai, Japan sits somewhere in between, but still on the higher end. Cooling alone is estimated to generate ~47,600 tons of CO₂ annually, largely because the grid depends heavily on LNG and coal. After the Fukushima nuclear disaster, much of Japan’s nuclear capacity was taken offline, which pushed the country toward imported fuels and made it harder to bring emissions down.

Singapore (2nd facility) is where the tradeoff becomes the most obvious. It has the highest PUE (1.145) and one of the most expensive electricity rates (~\$0.265/kWh), which pushes cooling costs to the top of the group. The grid is ~95% natural gas, so emissions stay high. But unlike most non-EU regions, Singapore is at least starting to price carbon—around ~\$18.62/ton today. Still far below EU levels, but it’s a signal that emissions are beginning to show up as a real economic cost.



The cost of inefficiency in a dirty-grid data center adds up fast. A 0.01 improvement in PUE at Changhua County, Taiwan reduces cooling overhead by 7.7%. That alone saves about \$1.64M a year and cuts 5,563 tons of CO₂ – that’s equivalent to roughly what 265,000 mature trees (+20 years old) could produce. Now zoom out. Changhua produces 16x the CO₂ of Hamina, Finland. Hamina needs about 220,000 trees a year to offset its cooling emissions. Changhua needs 3.4 million.

At that point, it’s not really an engineering problem but more of a location problem. The cleaner the grid and the more efficient the facility, the lower the carbon cost. But today, neither of those factors is consistently priced into where these data centers get built.

PUE isn’t a perfect metric – but it’s the best public signal we have for how efficiently a data center uses power. So let’s use it as a benchmark.

If Google is the gold standard, the real question becomes: *what should “good” actually look like for other hyperscalers operating in different parts of the world?*

For the 30 facilities Google already reports, we have the answer. But what about places like Jakarta, Mumbai, or Riyadh—where the next wave of data centers is actually being built?

That’s where the regression model comes in. Using Google’s data across 30 facilities over four years, we can isolate how much of PUE is driven by climate. Wet bulb temperature and latitude explain about 25% of the variation, which is enough to give us a reasonable estimate of what a

Google-level operator should be able to achieve in any given city – even before construction starts. Think of it as the **efficiency floor** – the baseline performance you’d expect before a single dollar of CapEx is committed.

Here are the predicted PUEs for the top 10 growing data center markets. For readers who want to go deeper, the next section walks through the methodology in more detail.

Region	Predicted PUE	Cooling Overhead (MW)	Cooling Overhead (kWh/yr)	Electricity Price (\$/kWh)	Annual Cooling Cost	Grid Carbon Intensity (gCO ₂ /kWh)	Primary Energy Source (2024, % of grid)	Annual CO ₂ from Cooling (tons)	Local Carbon Price (\$/ton)	Current Carbon Cost	Carbon Cost Gap
Tokyo, Japan	1.101	10.1 MW	88.7M kWh	\$0.202	\$17.91M	483	LNG 35%, Coal 30%, Nuclear 10%	42,871	\$1.93	\$83K	–\$2.93M
Seoul, S. Korea	1.095	9.5 MW	83.3M kWh	\$0.114	\$9.49M	416	Coal 36%, LNG 29%, Nuclear 25%	34,598	\$6.45	\$223K	–\$2.21M
Riyadh, Saudi Arabia	1.127	12.7 MW	111.2M kWh	\$0.038	\$4.23M	692	Gas 60%, Oil 40%	76,977	\$0.00	\$0	–\$5.42M
Dubai, UAE	1.132	13.2 MW	115.6M kWh	\$0.058	\$6.71M	468	Gas 99%, Solar 1%	54,047	\$0.00	\$0	–\$3.80M
Taipei, Taiwan	1.118	11.8 MW	103.6M kWh	\$0.187	\$19.37M	635	Coal 45%, LNG 35%, Nuclear 10%	65,805	\$0.00	\$0	–\$4.63M
Singapore	1.133	13.3 MW	116.3M kWh	\$0.265	\$30.81M	499	Gas 95%, Solar 3%	57,975	\$18.62	\$1.08M	–\$3.00M
Bangkok, Thailand	1.132	13.2 MW	115.7M kWh	\$0.083	\$9.60M	555	Gas 56%, Coal 20%, Renewables 14%	64,155	\$0.00	\$0	–\$4.51M
Kuala Lumpur, Malaysia	1.132	13.2 MW	115.2M kWh	\$0.083	\$9.56M	604	Gas 51%, Coal 34%, Hydro 10%	69,626	\$0.00	\$0	–\$4.90M
Mumbai, India	1.130	13.0 MW	113.5M kWh	\$0.078	\$8.85M	707	Coal 72%, Gas 8%, Renewables 17%	80,276	\$0.00	\$0	–\$5.65M
Jakarta, Indonesia	1.135	13.5 MW	118.0M kWh	\$0.070	\$8.26M	680	Coal 60%, Gas 22%, Oil 9%	80,252	\$0.72	\$58K	–\$5.59M

These are the markets where the next wave of hyperscale investment is going. Every city in this table carries a carbon gap, ranging from about \$2.2M in Seoul to \$5.6M in Mumbai per 100MW facility, every year. The two cheapest places to build and operate, like Riyadh and Jakarta, also have some of the largest carbon gaps. Riyadh runs at about \$4.2M in annual cooling cost thanks to heavily subsidized electricity (~\$0.038/kWh). Jakarta isn’t far behind at ~\$8.3M. But both carry over \$5M in unpriced carbon cost each year.

So, what can actually be done?

Three things that could move the needle immediately:

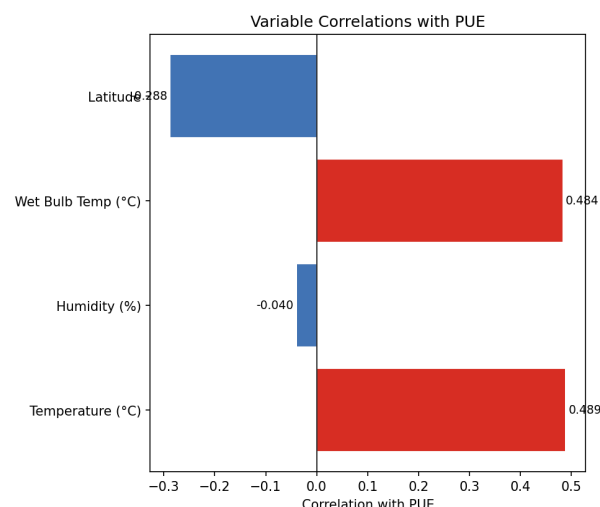
1. Require data centers to publish facility-level PUE and carbon disclosure. This entire analysis exists because Google publishes quarterly PUE data for every facility – AWS and Microsoft don't. Governments approving large data center builds, especially in Southeast Asia and the Middle East, should make facility-level energy and carbon disclosure a condition of operating. Because you can't hold companies accountable for what they don't have to report.

2. Price carbon where it actually matters. The EU Emissions Trading System works because it makes emissions show up in the cost of electricity. Power plants pay. That cost flows through. Large users like data centers feel it. The problem is that most fast-growing data center markets don't have anything similar. Which means companies can build in carbon-heavy regions without paying for the emissions that come with it.

3. Extend CBAM principles to digital infrastructure. The Carbon Border Adjustment Mechanism already does this for physical goods. If steel is produced in a low-carbon-price country and sold in Europe, the importer pays the difference. That same logic applies to compute. Right now, a data center in a low-carbon-price region can serve users in high-carbon-price markets with no adjustment. Extending CBAM to digital services is complex but even partial adoption would push companies to disclose more, build more efficiently, and think twice about treating low-carbon-price regions as a hidden subsidy

Regression Analysis

To estimate Power Usage Effectiveness (PUE), I evaluated 426 quarterly observations across 30 Google facilities from 2022–2025. The goal here is to understand how much of PUE is driven by external factors like climate, rather than engineering. I tested four observable variables: 1) temperature, 2) humidity, 3) wet bulb temperature, and 4) latitude



A. How each variable correlates with PUE

Temperature ($r = +0.489$) shows the strongest relationship with PUE. Warmer environments require more cooling, which increases total energy use relative to computing load. As temperature rises, PUE tends to rise with it.

Wet Bulb Temperature ($r = +0.484$) is nearly as strong. This metric captures both heat and humidity in a single value and reflects how effective evaporative cooling can be. When

conditions are both hot and humid, cooling efficiency drops and PUE increases.

Latitude ($r = -0.288$) is negatively correlated with PUE. Locations further from the equator tend to have cooler climates, which reduces cooling demand and lowers PUE. It acts as a broad geographic proxy for climate.

Humidity standalone ($r = -0.040$) shows almost no relationship with PUE. This is because humidity only becomes relevant when combined with high temperatures—it doesn't meaningfully affect cooling in colder environments.

B. Model selection

To identify the strongest predictors, I tested six models: two single-variable models and four two-variable combinations.

Models were evaluated using:

- adjusted R^2 (model fit adjusted for complexity)
- cross-validation R^2 and standard deviation (stability across samples)
- RMSE (average prediction error)
- AIC (fit relative to number of variables)
- p-values (statistical significance)
- VIF (multicollinearity check)

For readers less familiar with these metrics: the goal is simply to find a model that explains the data well without overfitting or using redundant variables.

One important constraint: temperature and wet bulb temperature were not used together in the same model. Their correlation is 0.977, meaning they capture almost the same information. Including both would introduce multicollinearity and distort the results. Humidity was also excluded as a standalone variable due to its near-zero correlation.

Here's the full model comparison:

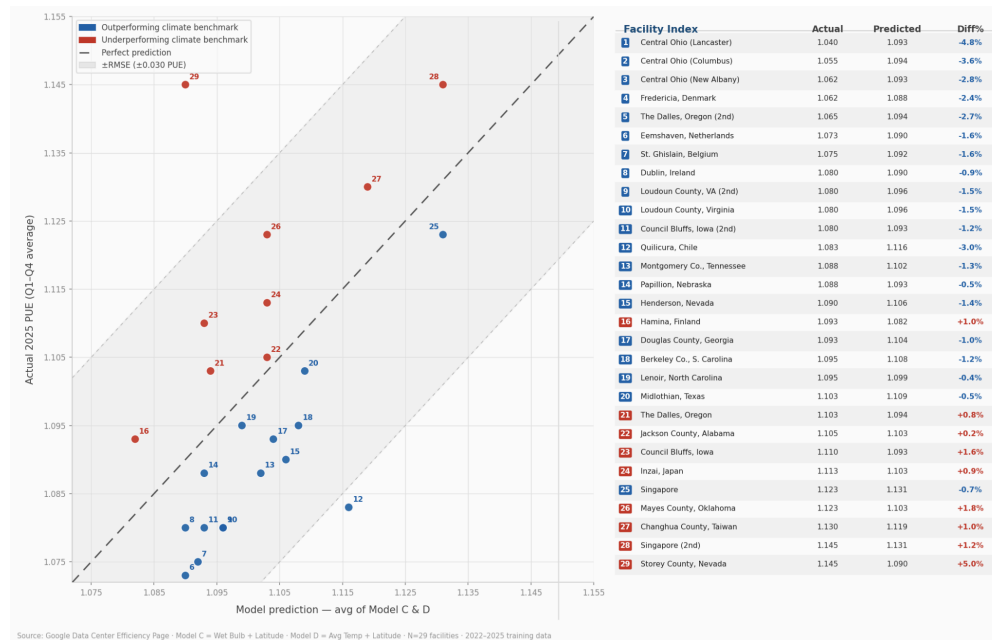
Model	Variables	Adj R ²	CV R ² (mean)	CV R ² (±std)	CV RMSE	AIC	Sig. p-value?	VIF	Key Issue / Notes
SINGLE VARIABLE MODELS									
A	Wet Bulb Temp	0.232	0.251	±0.153	0.0298	-1764.3	Yes	N/A	Statistically indistinguishable from Model C on CV R ² (0.251 vs 0.252) but misses the independent signal latitude adds.
B	Avg Temp	0.237	0.250	±0.125	0.0298	-1767.3	Yes	N/A	Marginally better Adj R ² but statistically indistinguishable from A. Loses humidity signal. Easier to explain to a general audience.
TWO VARIABLE MODELS									
C★	Wet Bulb + Latitude	0.241	0.252	±0.137	0.0298	-1768.3	Yes	1.6, 1.6	Best CV R ² of two-variable models (0.252). Both variables are significant (wet bulb p<0.001, latitude p=0.015). Clean VIF of 1.6.
D★	Avg Temp + Latitude	0.247	0.254	±0.112	0.0297	-1771.8	Yes	1.9, 1.9	Best AIC and Adj R ² . Lowest CV std dev — most consistent across folds. Latitude significant (p=0.011). Cleanest analytical model.
F	Wet Bulb + Humidity	0.242	0.242	±0.120	0.0299	-1768.9	Humidity: p=0.011 (marginal)	3.2, 3.2	Humidity coefficient near-zero (-0.00028). Wet bulb already encodes humidity — adding it separately is redundant. AIC doesn't justify inclusion.
G	Avg Temp + Humidity	0.236	0.235	±0.127	0.0301	-1765.3	Humidity: p=0.874 (not sig.)	3.6, 3.6	Humidity p-value = 0.874 — no statistical significance. Coefficient = 0.000017, effectively zero. Weakest two-variable model. Humidity eliminated.

Across all variations, the result is pretty consistent: climate alone explains about 23–25% of the variation in PUE across Google's fleet. roughly 23–25% of PUE variation across Google's fleet. That means the majority – roughly 75% – comes from everything else like cooling system design, hardware generation, AI-assisted energy optimisation, facility age, and operational decisions. In other words, climate sets the baseline. Engineering determines how far you can push beyond it.

After 5-fold cross-validation, the model shows an average prediction error of **±0.030 PUE units**. In practical terms, that means: if the model predicts a PUE of 1.10, the actual value is usually somewhere between ~1.07 and 1.13. So this isn't precise enough to design a facility around. But it *is* precise enough to answer a more important question: Is this location fundamentally easy or difficult to cool?

C. Why two models are retained

Both Model C (Wet Bulb + Latitude) and Model D (Temperature + Latitude) end up performing almost identically on every statistical measure — CV R^2 of 0.252 vs 0.254, RMSE identical at 0.0298, and both are statistically significant. To give a more balanced estimate, the predicted PUEs and carbon cost estimates in the previous section use an average of Model C and D. Below is a scatter plot that shows the predicted PUE from Model C and D and how it compares to the actual 2025 PUE from Google's public efficiency page.



Most facilities land within $\pm 2\%$ of the model — which means climate alone, observed from publicly available data, gets you surprisingly close to what Google actually achieves on the ground. The outliers are where the real story is. The Ohio cluster sits 3–5% below the prediction line meaning Google is getting better efficiency than the climate would suggest, likely through newer facilities and more advanced cooling systems. Storey County, Nevada sits 5% above the prediction line meaning something operational is holding it back, even though a dry desert climate should be easier to cool.

That's ultimately what this model is. It's a location benchmark — a way of asking, before you build, what the climate alone will cost you in efficiency and carbon. And that answer matters more now than ever. The four largest tech companies are spending over \$370 billion on data center infrastructure in 2025 alone. Most of that CapEx is going into the markets in this table—the ones with the largest carbon gaps and the lowest carbon prices.

These aren't one-off builds. They're 20–30 year infrastructure commitments, on grids that won't decarbonize overnight, in places that haven't started pricing the externality.

The problem is: *no one's asking this question before the decision gets made.*

